The Centriolar Imperative

Young Stem Cell Transplantation Extends Lifespan by Mitigating Post-Translational Damage to Mother Centrioles — A Unifying Perspective and the Novel Relapse Prediction

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Conflict of Interest Statement: The author declares no competing interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Acknowledgments: The author thanks the peer reviewers for their constructive comments, which have significantly improved the manuscript.

Abstract

The repeated observation that transplantation of young stem cells extends both healthspan and lifespan across diverse organisms—from Drosophila to mice and humans—presents a persistent mechanistic puzzle. Prevailing theories of aging, focused on telomere attrition, epigenetic drift, or macromolecular damage, fail to provide a unified, cell-autonomous explanation for why young cells, specifically, confer such profound systemic benefits. This review synthesizes convergent evidence through the lens of the Centriolar Damage Accumulation Theory of Aging (CDATA). CDATA posits that the non-dividing mother centriole, a microtubule-organizing center and basal body, accumulates irreversible post-translational modifications (PTMs)—notably excessive polyglutamylation—across the lifespan of a stem cell. This damage impairs its core functions: mitotic fidelity, ciliogenesis, and consequent Hedgehog/Wnt signal transduction, leading to flawed niche sensing and a decline in self-renewal capacity. We argue that young stem cells possess mother centrioles with less PTM damage, thereby restoring precise division geometry and ciliary signaling upon transplantation. Critically, CDATA generates a unique, falsifiable **Relapse Prediction (P11)**: functional rescue by young cells is inherently temporary, with relapse kinetics dictated by the recipient cell's division count, not calendar time, due to the irreversible nature of PTM accumulation. This contrasts with theories predicting stable rescue. We review supportive evidence from Drosophila centrosome misorientation, murine muscle and hematopoietic stem cell (HSC) studies, and human clinical outcomes. We further discuss therapeutic strategies targeting centriolar deglutamylases (e.g., CCP1/AGBL5) and the implications for biobanking. CDATA offers a unified framework explaining the conserved efficacy of young stem cell transplantation and provides a roadmap for novel, mechanism-based anti-aging interventions.

Keywords: aging, stem cells, centriole, polyglutamylation, transplantation, rejuvenation, Cdc42, CASIN, ATF5, CCP1

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1. Introduction: The Persistent Puzzle of Youthful Transplantation

A cornerstone finding in modern geroscience is that systemic factors of youth can rejuvenate aged tissues, most famously demonstrated by heterochronic parabiosis. Conversely, and perhaps more fundamentally, the transplantation of young stem and progenitor cells directly into aged organisms consistently produces significant extensions in both healthspan and lifespan. This phenomenon has been replicated across species and tissue systems: from the rescue of degenerative muscle and brain function to the robust, systemic lifespan extension observed following young bone marrow transplantation in mice (Kovina et al., 2013; Lavasani et al., 2012; Leins et al., 2022).

While empirically robust, this phenomenon lacks a consilient mechanistic explanation. Telomere theory suggests young cells possess longer telomeres, yet telomerase-deficient mice still benefit from young hematopoietic stem cell (HSC) transplants, and the lifespan extension often outpaces simple telomere renewal kinetics. Epigenetic clock theories imply a resetting of methylation patterns, but do not explain the primary, cell-intrinsic advantage of a young nucleus in an aged cytoplasmic milieu. Theories focused on mitochondrial dysfunction or protein aggregates struggle to explain why the introduction of a small, often poorly engrafted population of young stem cells can catalyze such disproportionate, organism-wide rejuvenation.

The field thus requires a paradigm that identifies a fundamental, time-dependent, and functionally critical lesion that is (a) inherently worse in old stem cells, (b) not easily reversed by the aged systemic environment, (c) directly responsible for declining stem cell function, and (d) diluted or bypassed by the introduction of young cells. We propose that the Centriolar Damage Accumulation Theory of Aging (CDATA) provides such a framework (Tkemaladze, 2023, 2026). CDATA identifies the mother centriole—a singular, non-renewable organelle that acts as the blueprint for cell division and ciliary signaling—as the central accumulator of irreversible damage. This review will articulate how CDATA mechanistically explains the lifespan-extending effects of young stem cell transplantation, synthesize convergent evidence from model organisms to humans, and introduce a novel, testable **Relapse Prediction** that distinguishes CDATA from other theories of aging.

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2. The CDATA Framework: Axioms, Irreversibility, and the PTM Mechanism

CDATA is built upon three core axioms derived from cell biology:

Axiom 1 (Asymmetry): During asymmetric stem cell division, the mother centriole is selectively retained by the self-renewing stem daughter cell. The daughter centriole is inherited by the differentiating cell.

Axiom 2 (Non-Renewal): Centrioles are not synthesized de novo in mammalian cells; they are templated from existing ones. The mother centriole is never rebuilt; its core structure persists across the stem cell lineage.

Axiom 3 (Function): The mother centriole serves two irreplaceable roles: (i) as part of the centrosome, it anchors the mitotic spindle, dictating division plane and symmetry; (ii) as the basal body, it nucleates the primary cilium, a critical signaling hub for pathways like Hedgehog (Hh) and Wnt.

The critical implication of Axioms 1 and 2 is the **¬R (No Reset) Argument**: the mother centriole is a permanent, lineage-spanning organelle that cannot be “rejuvenated” through normal cellular processes. Unlike nuclear DNA (repaired), mitochondria (fused and turned over), or even epigenomes (partially reprogrammable), the mother centriole’s structural core is subject to the cumulative wear and tear of a stem cell’s lifetime.

The primary mechanism of this wear is the accumulation of irreversible **post-translational modifications (PTMs)** on the long-lived tubulin polymers of the centriolar microtubules. A key culprit is **polyglutamylation**—the addition of glutamate side chains to tubulin. While moderate glutamylation regulates motor protein recruitment and ciliary assembly, excessive polyglutamylation, driven by time and enzymatic activity (e.g., TTLL family enzymes), leads to steric hindrance, destabilization of binding partners, and ultimately **ciliary loss and centrosome disorientation** (Tkemaladze, 2023). This damage accrues as a function of cellular age and metabolic activity, not merely division count, though divisions may exacerbate it.

Molecular Support for the PTM Mechanism: Recent work has provided direct molecular evidence for the critical role of polyglutamylation in centriole structure and function. Madarampalli et al. (2015) demonstrated that ATF5 forms a characteristic 9-fold symmetrical ring structure in the inner layer of the pericentriolar material (PCM) surrounding the proximal end of the mother centriole. Importantly, ATF5 interacts with polyglutamylated tubulin (PGT) on the mother centriole and with PCNT in the PCM, forming a tripartite complex PGT-ATF5-PCNT that is essential for mother centriole-directed PCM accumulation. ATF5 depletion blocks PCM accumulation at the centrosome and causes fragmentation of centrioles, leading to the formation of multi-polar mitotic spindles and genomic instability (Madarampalli et al., 2015). This provides direct experimental evidence for the central postulate of CDATA: polyglutamylation on the mother centriole is required for structural integrity of the centrosome, and its disruption leads to the very phenotypes CDATA associates with stem cell aging.

Further support comes from microinjection studies using the GT335 monoclonal antibody, which specifically recognizes polyglutamylated tubulin. Bobinnec et al. (1998) showed that GT335 injection causes complete disappearance of the centriole pair within 12 hours, accompanied by scattering of PCM and loss of the centrosome as a defined organelle. The authors concluded that **"a posttranslational modification of tubulin is critical for long-term stability of centriolar microtubules"** (Bobinnec et al., 1998, p. 1588). Notably, centriole recovery was observed to occur through synthesis of new centrioles rather than repair of old ones, supporting the ¬R argument that damaged mother centrioles are not repaired but must be replaced.

Thus, an aged stem cell carries a damaged "navigation and communication" organelle. It divides with misoriented spindles, failing to correctly apportion fate determinants or position itself within the niche. Its impaired cilium receives flawed Hh/Wnt signals, corrupting its interpretation of the microenvironment and leading to depletion, senescence, or malignant transformation. A young transplanted stem cell, carrying a mother centriole with minimal PTM damage, reintroduces a functional blueprint for precise division and accurate niche sensing, thereby restoring tissue homeostasis and systemic vigor.

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3. Convergent Evidence Across Organisms

The following evidence, summarized in Table 1, demonstrates age-related centriolar/centrosomal dysfunction and its rescue by young cells or targeted interventions, aligning with the predictions of CDATA.

3.1. Drosophila melanogaster

In the Drosophila testis, germline stem cells (GSCs) attach to the niche via centrosomally anchored adherens junctions. Cheng and Yamashita (2008) demonstrated that the centrosome (the fly equivalent of the centriole pair) in GSCs becomes misoriented with age, reducing the rate of symmetric, self-renewing divisions. This is consistent with the CDATA prediction of age-related centrosomal dysfunction leading to failed niche engagement and depletion of the stem cell pool. Notably, the authors observed that cell cycle arrest was transient, and GSCs appeared to re-enter the cell cycle upon correction of centrosome orientation (Cheng & Yamashita, 2008). This does not contradict CDATA, as it does not exclude the possibility that a subpopulation of cells with irreversibly damaged centrioles exists; rather, CDATA posits that the most severely damaged cells may be eliminated rather than repaired.

3.2. Mus musculus (Muscle System)

Lavasani et al. (2012) showed that aged muscle-derived stem/progenitor cells (MDSPCs) exhibit profound functional decline. Strikingly, co-culture with young MDSPCs, or transplantation of young MDSPCs into aged mice, rescued the function of the old cells in vivo and improved regeneration and strength. The authors noted that transplanted MDSPCs improved degenerative changes and vascularization in tissues where donor cells were not detected, suggesting that their therapeutic effect may be mediated by secreted factor(s) (Lavasani et al., 2012). These findings are compatible with CDATA, as secreted factors from young cells may transiently support the microenvironment while young cells also directly contribute progeny with functional centrioles.

3.3. Mus musculus (Hematopoietic System)

This system provides the most compelling lifespan data consistent with CDATA predictions.

Lifespan Extension: Kovina et al. (2013, 2019) performed syngeneic transplantation of young bone marrow (BM) into unirradiated, physiologically aged mice. With approximately 28% chimerism, they observed a remarkable 34% increase in median lifespan. This benefit—a substantial engrafted population driving a major systemic effect—is compatible with CDATA's prediction that introducing stem cells with functional centrioles can reset tissue-level organization.

Centrosome Polarity as a Target: Florian et al. (2020) identified loss of Cdc42 activity and associated centrosome polarity in aged HSCs. Treatment with CASIN, a Cdc42 activity inhibitor, restored centrosome polarization, reduced HSC aging phenotypes, and extended lifespan in transplantation models. The interpretation of these findings is compatible with CDATA, as centrosome polarity can be viewed as a readout of mother centriole functional integrity.

Functional Rejuvenation: Leins et al. (2022) extended this work, showing that ex vivo CASIN treatment of aged HSCs prior to transplantation not only improved hematopoiesis but also conferred significant lifespan and healthspan benefits, including improved muscle function in recipients. This demonstrates that targeting the centrosome polarization axis is sufficient for systemic rejuvenation.

Division History is Key: Shen et al. (2025) provided critical support for the ¬R argument. They found that within the old HSC pool, the CD150low subset (enriched for HSCs that have undergone fewer divisions) was significantly more potent in extending recipient lifespan (+9-12%) upon transplantation than the more frequently divided CD150high subset. These data are compatible with the CDATA prediction that less-divided HSCs have mother centrioles with less accumulated PTM damage, making them functionally "younger" irrespective of chronological donor age.

3.4. Homo sapiens (Clinical Evidence)

Bone Marrow Transplantation (BMT): It is a clinical standard that donor age is a primary determinant of BMT success. Younger donors lead to better engraftment, lower graft-versus-host disease, and improved long-term survival. This aligns with the core premise of CDATA.

Cell Therapy Trials: Zhu et al. (2024) conducted a randomized controlled trial using umbilical cord-derived mesenchymal stem cells (UC-MSCs) to treat frailty. UC-MSCs, possessing extremely "young" centrioles, demonstrated significant improvements in physical performance and inflammatory biomarkers. This directly parallels the murine data, suggesting a conserved mechanism.

Table 1: Convergent Evidence Compatible with CDATA Predictions

Organism	Tissue	Key Phenomenon	Reference	PMID
Drosophila	Testis (GSCs)	Centrosome misorientation ↑ with age → division rate ↓	Cheng & Yamashita, 2008	18923395
Mouse	Muscle (MDSPCs)	Co-culture/ transplant of young cells rescues function of old cells	Lavasani et al., 2012	22129202
Mouse	Hematopoietic (HSCs)	Young BM transplant → +34% median lifespan (28% chimerism)	Kovina et al., 2013, 2019	23967009; 31160718
Mouse	Hematopoietic (HSCs)	CASIN restores centrosome polarity →	Florian et al., 2020	32755011

Organism	Tissue	Key Phenomenon	Reference	PMID
		lifespan extension		
Mouse	Hematopoietic (HSCs)	CASIN-rejuvenated aged HSCs → lifespan & healthspan extension	Leins et al., 2022	36581635
Mouse	Hematopoietic (HSCs)	CD150low (less-divided) old HSCs → +9-12% lifespan	Shen et al., 2025	39743633
Human	Hematopoietic	Donor age is key determinant of BMT outcome	Clinical Standard	—
Human	Systemic	UC-MSC RCT improves frailty in aged patients	Zhu et al., 2024	38679727

4. Unified Interpretation: Why Young Centrioles Are Superior

The convergent data find a parsimonious explanation under CDATA. The universal efficacy of young stem cells is compatible with the hypothesis that they transplant functional mother centrioles with lower levels of damaging PTMs like polyglutamylation.

1. **Improved Mitotic Fidelity:** A young mother centriole ensures proper centrosome maturation, spindle anchoring, and orientation. This allows for precise asymmetric divisions, correct placement of daughter cells relative to the niche, and faithful segregation of fate determinants. This is compatible with the misorientation seen in aged Drosophila GSCs and the polarization defects in aged murine HSCs.

2. **Restored Ciliary Signaling:** The mother centriole, as the basal body, nucleates a fully functional primary cilium. A cilium built from a less-damaged basal body efficiently transduces Hedgehog and Wnt signals. These pathways are paramount for stem cell self-renewal, quiescence, and niche communication. Aged stem cells with polyglutamylated centrioles likely have short, absent, or dysfunctional cilia, leading to a state of “niche

deafness"—they cannot correctly interpret maintenance signals, leading to differentiation or exhaustion.

3. **The Systemic Effect:** A population of stem cells with accurate "navigation" (division) and "communication" (signaling) can repopulate a niche and re-establish normative signaling gradients. This can suppress the aberrant behavior of surrounding aged host stem cells (via non-cell-autonomous effects) and restore systemic tissue homeostasis, explaining the dramatic lifespan impact from partial chimerism.

Molecular Support for the "Superior Young Centriole" Hypothesis: The ATF5-PGT-PCNT interaction identified by Madarampalli et al. (2015) provides a molecular mechanism for age-related centriolar dysfunction. If polyglutamylation accumulates with age, the binding of ATF5 to PGT on the mother centriole would be progressively impaired, leading to reduced PCM accumulation, centriole fragmentation, and ultimately genomic instability. Young stem cells, with lower levels of polyglutamylation, would maintain intact ATF5-PGT binding and thus preserve centrosome structural integrity.

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5. The Novel Relapse Prediction: Distinguishing CDATA from Competing Theories

CDATA's ¬R Argument leads to a unique and falsifiable prediction absent from other major theories of aging. We designate this **Prediction 11 (P11): The Relapse Prediction**.

Statement: *Functional rescue of an aged stem cell population by co-culture with, or transplantation of, young stem cells is inherently temporary under syngeneic (non-immunogenic) conditions. The relapse—the return to an aged functional phenotype—will occur with kinetics proportional to the cumulative division count of the rescued (host-derived) stem cells post-rescue, not to the calendar time elapsed. This relapse will be observable even in the complete absence of immune rejection or systemic "re-aging."*

Rationale: In CDATA, the young cells provide a transient functional boost by diluting the pool of damaged centrioles. However, the host's aged stem cells retain their irreversibly damaged mother centrioles. Upon being called to divide (e.g., after injury or during normal turnover), these cells will again execute faulty divisions and signaling due to their damaged organelle. Their functional decline is thus deferred, not cured. Since centriolar PTM damage is linked to both time and metabolic cycles, but its functional manifestation is most acute during division, the rate of relapse will be driven by the proliferation demands placed on the host stem cell compartment.

Refinement of P11: Relapse is predicted to occur after multiple rounds of cell division, not immediately upon removal of the rejuvenating factor. Therefore, experiments showing CASIN effect stability for 6-24 hours after washout (Leins et al., 2022) do not test this prediction, as this timeframe is insufficient for multiple HSC divisions. Long-term serial transplantation experiments are required to test P11.

Contrast with Other Theories:

Theory	Predicted Outcome of Young Cell Transplantation
Telomere Theory	Stable rescue—young cells provide telomerase or long telomeres
Epigenetic/Programmed Theories	Permanent reprogramming—stable rejuvenation
ROS/Damage Accumulation Theories	Gradual improvement via reduced oxidative stress
CDATA (P11)	**Temporary rescue—relapse after multiple divisions**

Experimental Test Design (Co-culture Arm RELAPSE): A definitive test involves a two-phase, in vitro co-culture system using syngeneic young and old stem cells (e.g., HSCs).

1. **Phase 0 (Baseline):** Quantify functional deficit of old cells vs. young cells (e.g., colony-forming potential, spindle orientation).
2. **Phase 1 (Rescue):** Co-culture old cells with young cells (or in young cell-conditioned medium). Document functional rescue of the old cell population.
3. **Phase 2 (Relapse):** Physically or genetically (e.g., via a fluorescent marker) remove the young cells from the system. Continue culturing the now-isolated, previously “rescued” old cells.
4. **Prediction:** The old cells will relapse to their pre-rescue dysfunctional state within a predictable number of population doublings in Phase 2.

This prediction is a direct consequence of the non-renewable nature of the mother centriole and is a strong discriminator for CDATA.

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6. Therapeutic Implications: From Mechanism to Intervention

CDATA not only explains past data but also directs novel therapeutic strategies, moving beyond simply transplanting young cells.

1. **Targeting Centriolar PTMs Directly:** The upstream target is not Cdc42 (as with CASIN) but the tubulin deglutamylase enzymes CCP1 (AGTPBP1) and AGBL5. Pharmacological or genetic strategies to boost deglutamylase activity in aged stem cells could actively reverse the primary PTM damage, offering a more direct intervention than modulating downstream polarity effectors.

   Molecular Rationale: CCP1 (Agtpbp1) is a metallocarboxypeptidase that mediates deglutamylation of target proteins, including tubulins. Defects in Agtpbp1 cause Purkinje cell degeneration (pcd), with the molecular cause of neurodegeneration attributed to accumulation of glutamylation (InnateDB, 2014). Pan et al. (2025) demonstrated that CCP1 deficiency causes abnormal tubulin polyglutamylation, leading to shortened primary cilia in bone marrow mesenchymal stem cells (BMSCs). Importantly, reducing elevated polyglutamylation restored osteogenic differentiation and primary cilia length (Pan et al., 2025). These data establish that modulating deglutamylase activity directly impacts cellular function and suggest that enhancing CCP1 activity could be a viable therapeutic strategy for reversing age-related centriolar damage.

2. **Synergy with Partial Reprogramming:** Cellular reprogramming (OSKM) may, among its many effects, transiently reactivate deglutamylase expression or promote centriolar remodeling. Combining epigenetic reprogramming with targeted deglutamylase activation could be a powerful synergistic strategy.

3. **Personalized Bone Marrow Banking:** CDATA provides a strong theoretical foundation for the cryopreservation of one’s own bone marrow or HSCs at a young adult age (e.g., 20-30 years). This biobank would preserve stem cells with minimally damaged centrioles. Autologous transplant in late life would provide a source of functionally young stem cells, free of immunogenic risk, to reboot the hematopoietic and immune systems, potentially delaying age-related diseases and extending healthspan. The work of Shen et al. (2025) suggests that even within an older individual, identifying and expanding the least-divided (CD150low-like) stem cell fraction could have therapeutic benefit.

4. **Biomarker Development:** Quantifying centriolar polyglutamylation (e.g., via specific antibodies such as GT335 or through proteomics) in easily accessible cells could serve as a novel “centriolar age” biomarker, predictive of stem cell functional reserve and response to rejuvenation therapies.

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7. Limitations and Future Directions

While CDATA provides a compelling framework, several limitations must be acknowledged:

Absence of Direct Measurements: Direct measurement of polyglutamylation on mother centrioles of aged HSCs has not yet been performed. This remains a key gap that must be addressed through immunofluorescence with GT335 or similar antibodies coupled with super-resolution microscopy.

CASIN Mechanism: Whether CASIN affects centriolar PTMs remains to be determined. Current data indicate that CASIN acts as a selective inhibitor of Cdc42 (IC50 = 2 μM) and does not bind to RhoA, Rac1, RhoJ, RhoU, or RhoV (Leins et al., 2022). While CASIN treatment restores cell polarity, its effect on centriolar polyglutamylation has not been measured.

Age-Related Changes in ATF5-PGT Interaction: Age-related changes in the ATF5-PGT interaction have not been studied. Whether the binding affinity between ATF5 and polyglutamylated tubulin decreases with age remains an open question for future investigation.

Interpretation of GT335 Data: Recovery of centrioles after GT335 injection occurs through synthesis of new centrioles rather than repair of old ones. This is compatible with CDATA, as it supports the distinction between “repair” and “replacement” of centrioles, but it must be clearly articulated (Bobinnec et al., 1998).

Role of the Niche: Aging of stem cells also depends on the microenvironment (niche), including age-related decline in osteopontin (OPN) in the bone marrow stroma (Guidi et al., 2017; Guidi et al., 2021). CDATA focuses on cell-autonomous mechanisms, which is its strength, but this limitation should be recognized.

Call to Action: We call upon the geroscience community to prioritize direct measurement of centriolar polyglutamylation in aged HSCs using GT335-based immunofluorescence coupled with super-resolution microscopy. Such measurements would provide the definitive test of CDATA’s central prediction that PTM accumulation on mother centrioles is a primary driver of stem cell aging.

Recommended Experiments for Future Work:

1. Measurement of polyglutamylation on centrioles of HSCs from mice of different ages (IF with GT335, super-resolution microscopy)
2. Measurement of CCP1/AGBL5 activity in old vs. young HSCs (qPCR, western blot, enzymatic assay)
3. Hyperexpression of CCP1 in old HSCs—functional recovery?

4. Analysis of ATF5-PGT interaction in old vs. young HSCs—is it disrupted with age?
5. Measurement of PTM after CASIN treatment—does CASIN affect polyglutamylation?
6. Long-term serial transplantation experiments to test P11

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8. Conclusion

The Centriolar Damage Accumulation Theory of Aging provides a powerful, unifying framework that explains the decades-old observation that young stem cell transplantation extends lifespan. By identifying the mother centriole as a non-renewable organelle accumulating irreversible PTM damage, CDATA offers a parsimonious explanation for why youth is cell-intrinsically superior in a manner that telomeric, epigenetic, and damage-based theories cannot fully address.

The convergent evidence from flies to humans is striking: age-associated centrosome misorientation, the profound lifespan impact of young bone marrow, the therapeutic efficacy of centrosome-polarizing drugs like CASIN, and the superior function of less-divided HSCs all align under the CDATA umbrella. The theory's strength is further demonstrated by its capacity to generate unique, falsifiable predictions, most notably the Relapse Prediction (P11), which offers a clear experimental path to validate or refine the model against competing paradigms.

Acknowledging current gaps is crucial. Direct, high-resolution measurement of PTM accumulation (e.g., polyglutamylation) on the mother centrioles of aged versus young HSCs in vivo remains to be performed. This is the definitive experiment needed to solidify the mechanism. Furthermore, the precise link between specific PTM patterns and the loss of specific binding partners (e.g., Ninein, Cep164, ATF5) in aging stem cells needs elucidation.

If validated, CDATA shifts the gerotherapeutic paradigm. Instead of focusing solely on nuclei, epigenomes, or mitochondria, we must also consider the centriolar lifecycle. Interventions aimed at maintaining centriolar integrity—through deglutamylase activation, youthful biobanking, or division-rate management—could preserve the foundational geometry and signaling capacity of our stem cells, thereby sustaining tissue homeostasis and extending healthspan. The transplantation of young cells works because, in essence, it transplants young centrioles. The future of longevity medicine may lie not only in making old cells feel young but in ensuring their core architectural blueprint remains intact.

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